An Experimentalist Approach to Ethical Governance of Emerging Technologies*

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***Abstract*—The development of new technologies cannot be separated from the ethics of science and technology. The ethics of science and technology is embodied in the ethical spirit internalized in research and innovation activities on the one hand, and in the concrete rules of governance on the other. Only by combining these two aspects based on the spirit of experimentation is it possible to find an appropriate framework for the ethical governance of emerging science and technology. This paper reflects on the social responsibility of scientists, clarifies the nature of ethics of science and technology, and proposes the idea of "experimental space plus limited liability" for the governance of emerging ethics of science and technology and five principled considerations.**

Keywords—emerging technology · responsible research and innovation · experimental spaces · ethical governance

I. Introduction

The concept that "science and technology are the primary productive force" has deeply taken root in people's minds [1]. However, it's equally important to recognize that when science and technology are not developed and applied in a proper manner, they can turn into the primary destructive force. Consequently, the question of how to prevent science and technology from becoming the primary destructive force is a critical issue that demands our serious contemplation. Currently, this issue is particularly prominent in the risk management of emerging technologies, notably those represented by nanotechnology, biotechnology, cognitive science, psychological technology, robotics and artificial intelligence [2]. Given their novelty, emerging technologies is extremely challenging to foresee whether they will bring about benefits or disasters. As a result, it becomes arduous to make proper judgments and pinpoint potential risks.

Since emerging technologies cannot be precisely defined from the very beginning, innovators are bound to face risks that can never be entirely eradicated. With the development of these technologies, they interact non-linearly with diverse economic and social factors. Being continuously translated and put into practice by multiple stakeholders [3], they will inevitably give rise to unforeseeable economic and social outcomes, which may include risks that are difficult to control [4]. To cope with these new technologies, there are two extreme approaches. One is to let them develop freely and only take governance measures after negative consequences have emerged, when the damage has already been done. The other is to closely monitor them from the start and impose overly strict regulations, which can stifle their development. This situation is known as the Collingridge Dilemma [5]. There has been a great deal of researches on how to get out of this dilemma. Responsible research and innovation (RRI) is an approach that has gained considerable popularity in recent years [6].

However, the RRI framework encounters certain issues, with the most severe one being an inherent contradiction between the inclusiveness of stakeholders and the agility of collective action. Merely incorporating external parties (e.g., ethicists, the public) to some degree as insiders is insufficient to fundamentally resolve the problem [7]. This paper re-considers the nature of ethics of science and technology and puts forward an experimentalist governance framework for emerging technologies. This framework is structured around five guiding principles, aiming to both promote theoretical debates and establish actionable governance paradigms.

II. Wiener Vs. Bridgeman

Intrinsically, the ethical governance of emerging technologies is closely intertwined with the matter of the social responsibility of scientists and engineers since play a pivotal role in the advancement of these technologies. The ideal of the traditional scientist is science for science's sake, and thus has little, if any, social responsibility in developing science itself. Similarly, the primary professional obligation of traditional engineers centered on unwavering loyalty to their employers. Their fundamental responsibility lay in applying technical expertise to fulfill externally defined objectives through systematic problem-solving, a framework that inherently limited their perceived social accountability [8]. However, with the rapid development of science and technology and their increasingly farreaching influence on the economy and society, both scientists and engineers can no longer evade the issue of social responsibility.

In the late 1940s, the discussion on the social responsibility of scientists and engineers became a hot topic in the face of the devastating impact of military technologies, represented by nuclear weapons, on human society. In a letter to *The Atlantic Monthly in 1947*, "A Scientist Rebel," Norbert Wiener, the founder of cybernetics and a professor at MIT, responded to an

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inquiry from a Boeing engineer working on missiles, explicitly refusing to discuss technical issues with him and vowing "not to publish any future research that might cause harm through irresponsible military personnel" [9]. Wiener, acutely aware of the potential social ramifications of cybernetics, which was then an emerging technology, earnestly urged scientists and engineers to take into account the social outcomes of their work.

In response to Norbert Wiener's stance on research ethics, Harvard physicist Percy Williams Bridgman offered a pointed counterargument through his essay "Scientists and Social Responsibility". He rejected the notion of unilateral moral obligations for researchers, arguing that scientists should resist externally imposed responsibilities which he condemned as self-compromising and detrimental to professional dignity. He further contended that assigning ethical accountability to individual researchers fundamentally contradicts the principles of specialized labor division, a cornerstone of modern scientific progress. He emphasized that society grants scientists privileged status precisely because their discoveries require intellectual autonomy, and that demanding personal liability for societal applications of their work would undermine this foundational contract between science and civilization [10].

Wiener's position is surely respectable, but it is untenable to assume that scientists can always foresee the social consequences of their work. The indeterminacy of scientific discoveries' societal impacts stems from their embeddedness in political contestation and cultural negotiation, terrains where scientists possess neither methodological competence nor institutional mandate to intervene. In this regard, neither the scientific community nor individual scientists can be held responsible in any direct way for the social consequences of their activities [11]. However, this does not mean that the issue of the social responsibility of scientists should be ignored, as Bridgman might imagine it to be. We need to focus on the socio-political process around emerging technologies and interactions among the various actors involved in such a process. In this way, the core question would shift. It's no longer "What are the specific social responsibilities of scientists and engineers?", a question that implicitly assumes scientists should be accountable for the outcomes of their work. Instead, the key query now is "What can scientists and engineers contribute to the socio-political processes that govern the application of emerging technologies?" By getting involved in these processes, scientists and engineers can help shape policies, regulations, and public attitudes towards emerging technologies, ensuring that their development and application are in line with broader social values and interests. Therefore, participating in public understanding of science and technology and in policy debates is not just an option but a fundamental obligation for scientists and engineers.

Currently, humanity has stepped into the risk society in which science and technology, once regarded as the driving forces of progress, have now transformed into sources of various issues and have become the targets of public scrutiny [12]. In this context, the debate between Wiener and Bridgman remains instructive. Scientists and engineers bear a distinct ethical responsibility for the issues posed by emerging technologies. This responsibility is mainly reflected in the unique role of scientists and engineers as intellectuals, actively engaged in the socio-political process that shape the interactions among science, technology, and society.

Jean-Paul Sartre, a key 20th-century philosopher, redefined the intellectual's role and social responsibility. He argued that intellectuals, endowed with critical thinking, bear a unique duty to expose injustices and drive change. This obligation stems from their inherent freedom to act. Sartre called on intellectuals to transcend the realm of abstract theory and actively immerse themselves in society. He emphasized that they must balance universal concerns with local realities, offering practical and meaningful solutions to the communities they serve, and that courage is needed to resist power and challenge unjust authorities, while also caution against intellectual arrogance [13]. Sartre's views, strongly relevant in the mid-20th century, remain deeply insightful for today's intellectuals as they navigate their role and responsibilities in governance of emerging technologies.

III. The Nature of the Ethics of Science and Technology

Scientists and engineers bear certain social responsibility, which is an ethical imperative. However, the nature of ethics in science and technology is often misunderstood, perceived as a fixed dogma or a ready-made solution. After watching ethicists engage in debates that end without clear-cut conclusions, some of them may frequently head home disheartened and simply go about their regular duties. This situation reflects the current attitude of some scientists and engineers towards the ethics of science and technology.

As a matter of fact, ethics is fundamentally about societal concern and a sense of responsibility, rather than a fixed dogma or algorithm, and thus cannot straitforwardly provide specific answers. Scientists and engineers who actively participate in shaping ethical norms in science and technology are fulfilling their rightful role. By definition, ethics cannot function solely as external binding norms. It must also be internalized, becoming part of an individual's moral character. A person can only be considered to have "acted ethically" when they genuinely approval of ethical norms and act in accordance with them. If someone is forced to follow ethical norms without internalizing them, their actions are unlikely to be truly ethical. In this sense, to do good deeds, the first thing that matters is the good heart, and then the good deeds; if the good heart and the good deeds are united, that is the ideal state. Following the discovery of genetic recombination technology in the 1970s, biologist Paul Berg proactively suspended his experimental work convened peer scholars to discuss the implications, and formulated the relevant research specifications and guidelines, thereby fulfilling the social responsibility as a scientist [14]. Berg set an example of the role of a "responsible researcher". He demonstrated his sense of responsibility and took actions to translate this sense into concrete rules for future generations.

Ethical norms typically present themselves in a form that is overly general and abstract, rendering them insufficient for directly steering actions. To translate these norms into practical conduct, a process of interpretation becomes indispensable. In essence, adhering to ethical norms is a journey of "ethical interpretation." The inherent vagueness of these norms means that they cannot be fully understood or effectively applied in isolation. Instead, they must be contextualized and given concrete expression. This involves considering the specific

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circumstances, cultural backgrounds, and the diverse values at play in different situations. Only through such a nuanced approach can the true meaning and practical implications of ethical norms be unearthed, enabling individuals and organizations to make morally sound decisions and act in ways that are consistent with the spirit of these norms [15]. Therefore, it is very important for scientists and engineers to conduct appropriate "ethical interpretations", which mainly depend on their moral character, including moral integrity and professional competence, especially when assessing the risks of emerging technologies.

Given this, it is crucial that scientific communities or organizations engaged in innovation practices translate ethical norms into more explicit standards. This process of translation can be regarded as a specialized mode of ethical interpretation. For example, the chemical industry standard "Guidelines for the Implementation of Responsible Care" (HG/T4184-2011) issued by the Chinese National Standardization Administration Committee specifies 12 guiding principles for the implementation of Responsible Care and six guidelines for its implementation. Reflecting the concretization and institutionalization of ethical norms in the chemical industry, these guidelines delineate the actions that engineers and managers are expected to take in specific circumstances.

When it comes to emerging technologies, these specific rules typically have their genesis within the scientific community, or at the very least, demand its leadership or in - depth participation. Given that these rules are an inherent part of scientific and technological advancement, they are not set in stone. Instead, they must be continuously adjusted as technology progresses. It is quite acceptable for these rules to be in an immature state, as long as there is scope for iterative learning. In fact, it is a well-known fact that any new thing cannot be fully-fledged and mature from the start, and thus, they necessitate an inclusive environment. In this particular context, the ethical spirit is characterized by experimentation, learning, and exploration.

IV. A CASE: THE CHANGE OF THE "14-DAY RULE"

As an emerging technology, human embryo research raises significant ethical questions. The establishment and subsequent relaxation of the "14-day rule" for such research illustrate how ethical frameworks in science and technology evolve in response to new knowledge and societal values.

In 1978, the world's first in vitro fertilized baby was born in the United Kingdom. In 1979, the Ethics Advisory Committee of the U.S. Department of Health, Education and Welfare first recommended the "14-day rule". Following this, the British government assembled a group of experts to establish the Warnock Commission. In 1984, this commission issued a report that strongly recommended that human embryos created through in-vitro fertilization, regardless of whether they were frozen or unfrozen, should not be allowed to survive beyond the 14th day after fertilization if no embryo transfer took place within the first day of fertilization. At the same time, it is proposed to make it a criminal offence to experiment with any human embryo beyond this period [16]. This marked the formal establishment of the "14-day rule", which has since been adopted by numerous countries as a guiding principle for human embryo research..

The 14-day point was chosen primarily because the development of the human nervous system begins around 17 days after fertilization. Before day 14, embryos are not capable of feeling pain and do not present significant ethical concerns for experimental use. Additionally, technical limitations at the time made it practically impossible for human embryos to survive beyond 14 days in vitro. As a policy tool, the "14-day rule" created a defined space for experimentation while setting a clear, legally binding boundary. Without such a rule, neither an outright ban on embryo research nor an unrestricted use of embryos would be viable. An outright ban would deprive humanity of the potential benefits of embryo research, while unrestricted use would lead to endless ethical debates and even societal conflicts.

However, by 2016, two research groups from the UK and the US successfully allowed human embryos to develop in vitro to day 13 post-fertilization, terminating the experiment just before reaching the 14-day limit as mandated by the rule. It is entirely possible that they could have continued to grow embryos beyond this period if they had chosen to do so. Moreover, over the past few decades, the scientific community has developed increasingly complex models of human stem cell embryos, revealing new ways to study early human development. These advances require the scientific community and regulatory communities to reassess the validity of the “14-day rule” and consider redefining the ethical and practical boundaries of embryo research. In May 2021, the International Society for Stem Cell Research (ISSCR) released updated Guidelines for Stem Cell Research and Clinical Translation, formally relaxing the “14-day rule”. The new guidelines suggest that research on human embryos cultured for more than two weeks should be considered on a case-by-case basis, subject to several stages of review to determine when experiments should be halted. This change provides scientists with greater flexibility to explore human development and disease [17].

The history of the "14-day rule" shows that foundational guidelines are essential for research. However, rigid adherence to specific rules without adapting them as technology advances can also hinder progress and innovation. Flexibility and regular reassessment are crucial to ensuring that ethical and scientific frameworks remain aligned with the evolving capabilities of science and technology. The process of introducing and adjusting research guidelines involves dialogue among various stakeholders, but the focus of their roles can differ significantly. The question of how far to extend the "14-day rule" has now become open for discussion. Professionals are tasked with proposing viable options, beginning with the identification of a timeframe that could help address scientific or medical challenges. Ethicists then engage in further deliberation based on these professional recommendations, ensuring that their arguments align with contemporary ethical norms and legal frameworks. This integration is crucial for embedding the proposed changes into the existing regulatory system. Finally, the legalization process requires public participation, input from science and technology ethics committees, and legislative decisions. In orther words, the establishment of enforceable guidelines necessitates three key processes: professionalization, rationalization, and legalization. Only after these three processes can the rules be finalized.

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V. Experimentalist Approach to Ethical Governance

The history of the 14-day rule shows that translating ethical principles into governance rules enables experimentation in emerging technologies. These governance rules are inherently experimental and should adapt as technology evolves. This suggests that the ethical governance of emerging technologies requires balancing risk and development, as well as responsibility and action.

Before proposing an experimentalist approach to ethical governance of emerging technologies, it is necessary to critically examine the RRI framework. Emerging from Europe and the United States in recent years, RRI emphasizes addressing social, ethical, and legal issues in research and innovation; It also advocates for the joint participation and collective consultation of stakeholder groups, as well as early intervention and real-time assessment by ethicists and other humanists [18]. A typical RRI process includes four dimensions: anticipation, reflectivity, engagement, and feedback. The anticipation dimension combines scientific evidence with future analysis to help innovators better understand the opportunities and challenges they face; the reflectivity dimension encourages innovators to examine their own actions and the innovation process; the engagement dimension involves situating the vision, purpose, problems, and dilemmas within a larger social context to facilitate collective deliberation through participation; Finally, the feedback dimension involves timely input from stakeholder groups to guide the framework and direction of innovation activities [19].

While RRI offers a potential path to address the risks associated with emerging technologies, it also faces fundamental dilemmas. First, the inherent uncertainty of emerging technologies means that problems often arise not because actors are irresponsible, but because they lack the resources or means to act responsibly. Second, according to the operational model of responsible innovation, when problems persist, it becomes difficult to assign responsibility, as the outcomes are the result of collective participation. Third, if the "rules" for RRI were fully implemented, many innovations might become infeasible due to high transaction costs and the slow, ineffective nature of collective action involving numerous stakeholders. Essentially, the issues raised by emerging technologies extend beyond the discourse of responsibility. It's not the case that as long as all participants are responsible, there won't be any problems with emerging technologies. As a result, the question of how to ethically govern emerging technologies remains unresolved. Moreover, the RRI has often been enforced upon scientists and innovators from an external perspective, seemingly establishing a moral high ground for ethicists to "dictate" to practitioners. Consequently, it may undermine the efficacy of the approach.

To address these challenges, I propose an alternative governance approach: “gradually expanded experimental space plus limited liability”. The development of emerging technologies requires designated space for exploration, but this space must be limited initially to mitigate the significant risks that these technologies may pose [20]. It is essential to allow emerging technologies to develop and optimize their experiments rather than stifle them from the outset. As our understanding of associated risks improves, this experimental space can gradually expand, allowing innovators to conduct experiments with greater freedom while bearing their own risks. Advances in technologies such as physical and computer simulations have also expanded the experimental space for emerging technologies, significantly reducing costs and risks. As these technologies mature and risk assessments deepen, the experimental space can grow further, enabling more extensive testing. At a certain point, emerging technologies can be introduced to society with only limited responsibility, where mechanisms such as market competition, diversity, and lead users act as key filters. This process represents a form of learning through experimentation.

The construction of experimental spaces for emerging technologies depends on the development of supportive public policies. These policies should regulate the experimental space by permitting innovation while requiring limited liability from innovators. They should be primarily "negative" policies aimed at preventing disruptive mistakes and trading time for safety. Under this framework various stakeholders are part of an expanding process where they can engage in innovation and play a critical role in fostering a spontaneously expanding innovation order. Without such a framework, innovation as creative destruction would not be made possible. Certainly, how to create experimental spaces that embrace creative destruction and establishing an expanding experimental order remains a public policy challenge.

VI. Principles for for Navigating the Experimental Space

The development of new technologies cannot be separated from the ethics of science and technology. This ethical framework manifests in two key ways: first, through the ethical spirit guiding research and innovation activities, and second, as concrete rules and frameworks for governance. To establish an effective ethical governance framework for emerging technologies, these two aspects must be integrated through an experimental approach. This involves creating iterative spaces for experimentation in the development of emerging science and technology, allowing for continuous learning and adaptation. Based on this understanding, five fundamental principles for navigating the experimental space can be summarized as the following.

A. Principle 1: In the governance of emerging technologies, having rules is far better than having none at all

Rules provide a framework for the governance of emerging technology, which reduces uncertainty and risk to some extent, and is an embodiment of the ethical spirit. Without minimal rules, society risks devolving into a chaotic "jungle" in some degree, where individuals act without restraint, leading to increased uncertainty and risk.

The presence of rules creates structure and predictability allowing individuals to understand expected behaviors and standards [21]. This clarity enhances efficiency, fosters collaboration, and mitigates risks associated with ambiguity. Moreover, rules are often designed to prevent negative behaviors or accidents, thereby serving a crucial role in risk management of emerging technologies. Additionally, rules often reflect the core values and ethical standards of a society or organization. By adhering to these rules, individuals and

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groups demonstrate their commitment to these vales and standards, which reinforces a shared ethical spirit [22].

These rules can either be deliberately and consciously formulated by a collective or evolve spontaneously in practice [23] and then be legalized on a broader scale. Regardless of which of the above situations it falls into, it fully demonstrates the crucial significance of the minimum rules for social survival and the development of emerging technologies.

B. Principle 2: The formulation of rules requires rigorous argumentation that includes both scientific and philosophical perspectives

The formulation of rules for governance of emerging technologies must be grounded in both scientific and philosophical arguments. In one hand, scientific arguments rely on empirical evidence and data to demonstrate the rule's necessity and effectiveness. On the other hand, philosophical arguments explore the ethical and moral dimensions of the rule, ensuring that it aligns with societal values and principles of justice [24][25]. As a result of effective argumentation, well-justified rules are more likely to be accepted and complied with by the public as their rationale is clear. In contrast, rules without strong justification often encounter skepticism and resistance, potentially leading to their abandonment or ineffectiveness.

Fundamentally, argumentation elevates the rule-discussion by introducing higher-order principles. It rationalizes the relationship between the rule under examination and other co-level rules. It also precisely defines the contextual relevance and specific conditions for the rule's application. Additionally, a key part of argumentation is considering the potential consequences of implementing the rule, including both intended and unintended effects. By carefully evaluating these outcomes, policymakers can anticipate problems and adjust the rule accordingly.

Therefore, to create rules that are effective, sustainable, and just, we should ground them in scientific evidence, align them with ethical principles, ensure their contextual relevance, and factor in potential consequences. This comprehensive approach highlights the overall significance of argumentation in rule-making for governance of emerging technologies.

C. Principle 3 The introduction of rules should, in itself, be a process that adheres to pre-established regulations

Rules are meant to be followed, so introducing new rules should promote compliance and acceptance among those they apply to. New rules are more likely to be accepted and adhered to when grounded in established legislative frameworks that the public already recognizes and follows. This practice of basing new rules on existing norms is a legitimization mechanism, which effectively boosts acceptability and compliance.

Philosophers like John Rawls and Jürgen Habermas offer insights on rule-making. Rawls posits that justice principles are what free and rational individuals would accept in an equal starting point as the basis for their association [24]. Habermas emphasizes rational discourse and public-reasoned consensus in legitimizing rules [26]. In this sense, by grounding new rules in widely-accepted principles and ensuring transparent, inclusive processes, governance can make rules both legally binding and morally persuasive.

Certainly, procedural justice also matters significantly for rule acceptance. People are more likely to comply with rules when they believe the decision-making is fair and legitimate [27], which highlights the importance of public participation in rule-making, ensuring their voices are heard and rules meet their needs. This is also a fundamental requirement of a democratic society.

In short, the legitimization of rules through adherence to existing legislative frameworks and public engagement in the rule-making process not only enhances their acceptability but also increases the likelihood of compliance. By following these principles, governance can foster a culture of voluntary compliance and societal harmony.

D. Principle 4: It always makes sense to roll out conservative rules in the first place

Beginning with conservative rules aligns with the principle of prudence, an emphasis on caution and foresight in decision-making. Under conditions of high uncertainty regarding emerging technologies, this ensures that potential negative consequences are minimized before more expansive or liberal regulations on emerging technologies are introduced.

This governance strategy operationalizes Aristotelian phronesis [28] through phased regulatory implementation, where initial conservatism serves as both risk mitigation mechanism and trust-building infrastructure. By prioritizing precautionary measures in contexts of uncertainty [29], policymakers create epistemic scaffolding to systematically identify latent systemic risks while accumulating empirical evidence through controlled deployment. This aligns with Tom R. Tyler's findings on procedural justice, where caution enhances public perception of legitimacy and consequent compliance [27].

The calibrated restraint inherent in this approach paradoxically enables responsible innovation, as temporary constraints foster disciplined experimentation cycles that surface critical operational parameters. As empirical understanding crystallizes through iterative feedback loops, strategic relaxation of initial safeguards becomes ethically justifiable precisely because it emerges from demonstrated operational mastery rather than speculative optimism.

Ultimately, this dynamic model transforms prudence from static risk-avoidance into adaptive governance architecture that synchronizes technical feasibility with societal expectations, thereby converting the precautionary principle's theoretical mandate into actionable policy pathways that balance innovation velocity with democratic accountability.

E. Principle 5: There is no one-size-fits-all rule; any rule is experimental and must be continuously adjusted

The development of rules for emerging technologies is an iterative and evolutionary process, not a one-time, definitive act. This perspective is crucial for effective governance under the current context.

Due to the limitation of human cognitive abilities, it is impossible to formulate a perfect rule at the very start of emerging technology development. Instead, rules should be tentatively established and then continuously refined through practical experience and learning.

Moreover, the rapid advancements in science and technology, along with the dynamic nature of social life, demand that rules be adaptable. As Richard Florida pointed out in his work on the creative class, societal transformations

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require regulatory frameworks that can evolve in tandem with technological and cultural shifts [30]. If rules fail to adapt, they will become obsolete and ineffective. Therefore, the constant adjustment of rules related to emerging technologies is not only inevitable but also essential for their effectiveness.

This view aligns with the concept of adaptive management in policy-making, which emphasizes learning from ongoing experiences and adjusting strategies accordingly [31]. By acknowledging that rules are provisional and subject to modification, policymakers can foster a culture of flexibility and responsiveness, enabling them to address emerging challenges more effectively.

The iterative process of rule-making allows for the integration of new knowledge and insights. Amartya Sen's capability approach underscores the importance of continuous improvement in policies to enhance human well-being [32]. When policymakers view rules as works in progress, they can engage in a sustained dialogue with stakeholders, incorporating diverse perspectives and evidence-based findings to optimize regulatory outcomes.

Recognizing that no rule is static or infallible is crucial. Embracing an adaptive and experimental approach to rule-making enables societies to navigate complexity and change more adeptly, ultimately leading to better-informed and more responsive regulatory frameworks for governance of emerging technolgies.

VII. Conclusion

The development of new technologies cannot be separated from the ethics of science and technology. The ethics of science and technology serves as a critical bridge between technological advancements and societal well-being. It not only guides researchers and innovators in adhering to ethical principles but also ensures that technological progress aligns with the broader goals of social justice, human dignity, and sustainable development. At its core, the ethics of science and technology is rooted in values such as responsibility, transparency, fairness, and sustainability. These principles must be translated into actionable guidelines applicable across diverse fields of research, including artificial intelligence, biotechnology, robotics, and environmental science.

The ethical governance of emerging technologies necessitates a balance between risk and development, along with a proper equilibrium of responsibility and action. However, the current approach to governance of emerging technologies such as RRI has its limitations. To deal with this issue, I propose an alternative governance approach to governance of emerging technologies: “gradually expanded experimental space plus limited liability”. Rather than suppressing emerging technologies from the very beginning, it is vital to let them develop and refine their experiments. As we gain a better understanding of the related risks, the scope for experimentation can gradually widen. This enables innovators to carry out experiments more freely, all the while taking on their own risks. The establishment of experimental spaces for emerging technologies hinges on the formulation of supportive public policies. These policies should manage the experimental realm by encouraging innovation while holding innovators accountable to a limited extent.They should mainly be of a “negative” nature, with the primary objective of averting disruptive errors and sacrificing time to ensure safety.

To navigate this gradually expanding experimental space, we should adhere to five fundamental principles with experimental spirit: First, in the governance of emerging technologies, having rules is far better than having none at all. Second, the formulation of rules requires rigorous argumentation that includes both scientific and philosophical perspectives. Third, the introduction of rules should, in itself, be a process that adheres to pre-established regulations. Fourth, It always makes sense to roll out conservative rules in the first place. Fifth, there is no one-size-fits-all rule, and any rule is experimental and must be continuously adjusted.

The process of transforming abstract ethical principles into concrete guidelines requires the active involvement of multiple stakeholders, including researchers, policymakers, ethicists, and civil society representatives. This collaborative approach ensures that ethical norms are not only professionalized and rationalized but also legitimized through public discourse and consensus-building. It also facilitates the adaptation of ethical guidelines to the specific contexts and challenges of different research areas, making them more practical and applicable in real-world scenarios.

Ultimately, the integration of ethical norms into the development of emerging technologies is not merely a regulatory requirement but a moral obligation for innovators. By ensuring that their work is ethically sound and socially responsible, technologists can contribute to the creation of a future technological civilization that respects human rights, promotes societal well-being, and addresses global challenges such as climate change and inequality. In doing so, they fulfill their historical responsibility as architects of the technological future, shaping innovations that serve the greater good and uphold the dignity of all individuals. The ethics of science and technology must be rigorously defined, continuously refined, and systematically applied to research and innovation. Only through such an approach can ethical norms be effectively integrated into the governance of emerging technologies, ensuring that technological progress is both responsible and beneficial to society at large.

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